

Derbyshire Shared Care Pathology

Neutropenia in Adults

Definitions

- Reference Range 2.0 – 7.0 x 10⁹/L
- Mild Neutropenia 1.0 – 1.9 x 10⁹/L
- Moderate Neutropenia 0.5 – 0.9 x 10⁹/L
- Severe Neutropenia <0.5 x 10⁹/L

Introduction

Mild neutropenia is a common reason for referral to the Haematology clinic, but is unlikely to be of clinical significance in the majority of cases..

Causes

Transient

- Lasting less than 4 weeks. Normally mild. Usually related to viral infections and not associated with clinical problems. This type of mild neutropenia may occasionally persist for several months.

Persistent

- Duffy-null associated neutrophil count (DANC). (Previously termed Benign Ethnic Neutropenia; a neutrophil count down to 1.0 x 10⁹/L is common in individuals of Afro-Caribbean and Middle Eastern ancestry, it does not increase the risk of infections or require investigation).
- Autoimmune disorders, e.g. SLE and rheumatoid arthritis
- Splenomegaly
- Drug related (see British National Formulary and appendix to this guideline)
- B12 / folate deficiency
- Haematological disorders (e.g. myelodysplastic syndromes, leukaemias, lymphomas, myeloma)

The severity of neutropenia will depend on the underlying cause. Neutropenia secondary to ethnic variation or splenomegaly tends to be mild, whereas drug (e.g. chemotherapy) related neutropenia can be severe. Normally there is a temporal association between commencement of the drug and development of the neutropenia.

Patients with neutrophils <1 x 10⁹/L in association other cytopenia (Hb <100 g/L, platelet count <100 x 10⁹/L) should be referred urgently for outpatient assessment

What to do next

Please follow the Flow Chart on page 2.

Contacts

Urgent queries:

RDH Consultant Haematologist
CRH Consultant Haematologist

Via RDH switchboard
Via Consultant Connect or on-call Consultant
through CRH switchboard

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Reviewed by: Dr J Tam & Dr K Lam & Dr E Welch, Consultant Haematologists at CRH; Dr F Al Kaisi, Consultant Haematology UHDB, Dr P Blackwell GP Hannage Brook Medical Centre, Ms S Knowles, Consultant Clinical Scientist UHDB

Authorised by Julia Forsyth

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QHB Consultant Haematologist

Via QHB Switchboard

Non urgent queries:

RDH Consultant Haematologist

CRH Consultant Haematologist

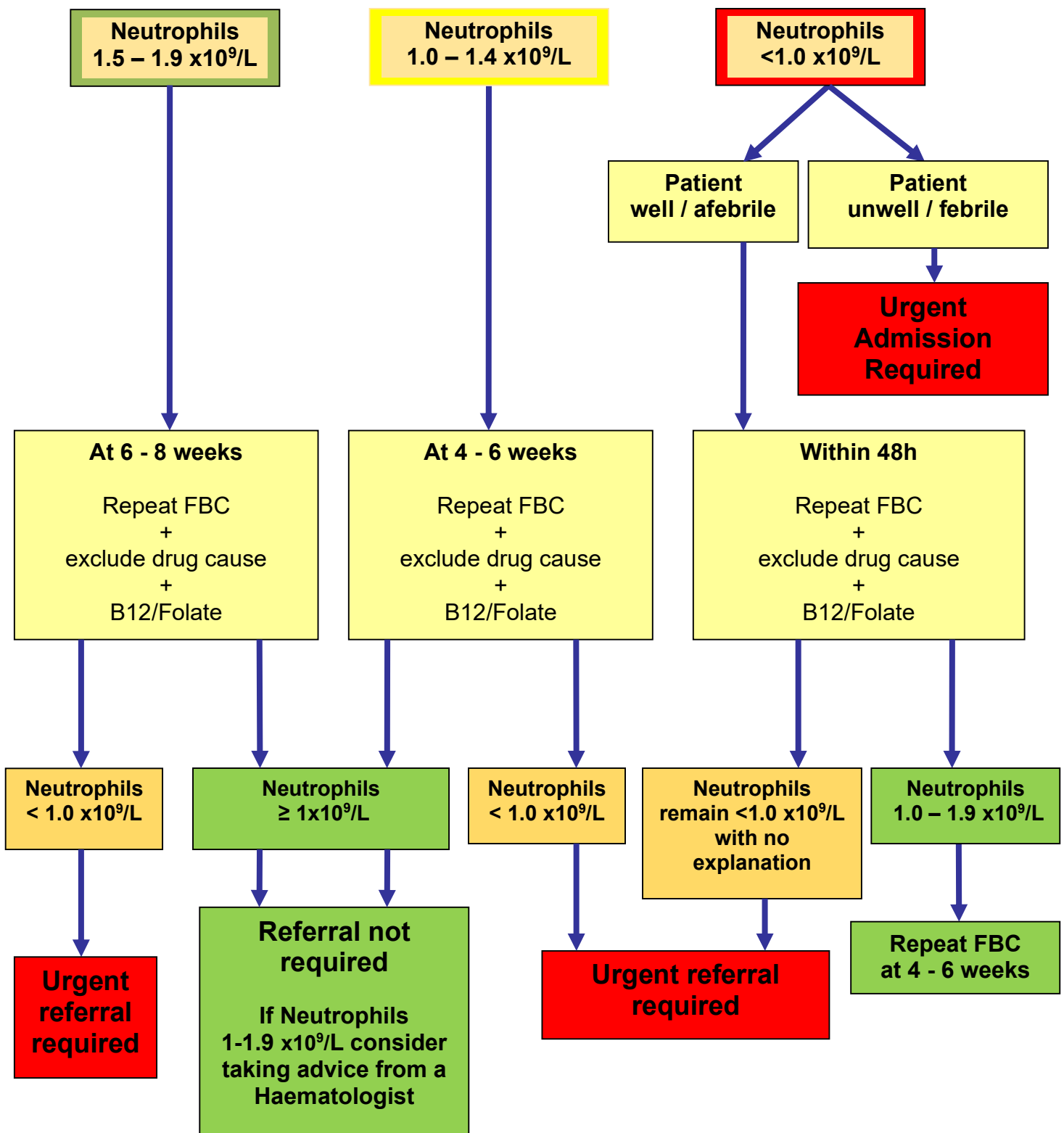
QHB Consultant Haematologist

01332 787973 (9am – 5pm)

crhft.cabadviceandguidance@nhs.net

Haematology secretaries 01283 593432 or extn
4025/4032 via QHB switchboard

Neutropenia Guidance Flowchart



Patients with neutrophils <1 x 10⁹/L in association other cytopenia (Hb <100 g/L, platelet count <100 x 10⁹/L) should be referred urgently for outpatient assessment

Appendix: List of common drugs known to cause neutropenia.

Class	Examples
Antimicrobials	Penicillin, cephalosporins, vancomycin, chloramphenicol, gentamicin, clindamycin, doxycycline, nitrofurantoin, minocycline, griseofulvin, metronidazole, rifampicin, sulfonamides, chloroquine, linezolid, hydroxychloroquine, dapsone, ciprofloxacin, trimethoprim, aciclovir
Analgesics	Ibuprofen
Antipsychotics, antidepressants	Phenothiazines, clozapine, diazepam, quetiapine, haloperidol
Anticonvulsants	Valproic acid, phenytoin, carbamazepine
Antithyroid drugs	Thiouracil, propylthiouracil, carbimazole
Cardiovascular drugs	Propranolol, thiazides, spironolactone, nifedipine, ticlopidine
Immunosuppressants	Methotrexate, mycophenolate, azathioprine, rituximab
Miscellaneous	Proton pump inhibitors, allopurinol, colchicine, bezafibrate, tamoxifen, penicillamine, retinoic acid, metoclopramide, ethanol, chlorpropamide, levodopa

Authors: Dr Chris Millar, January 2012		
Reviewed by:	Date:	Expiry date:
Dr C Millar, Dr P Blackwell, Mrs H Seddon	Aug 2014	31 st Aug 2016
Dr C Millar, Dr P Blackwell, Mrs H Seddon	Dec 2016	31 st Dec 2018
Dr R Faulkner, Dr K Lam, Dr E Welch, Dr I Amott, Dr P Blackwell, Mrs H Seddon	June 2020	30 th June 2022
Dr J Tam, Dr K Lam, Dr E Welch, Ms H Seddon	Oct 2022	31 st Oct 2024
Dr J Tam, Dr K Lam, Dr E Welch, Dr F Al Kaisi, Ms S Knowles	Oct 2024	31 st Oct 2026